# Enabling an operational, open and FAIR EOSC ecosystem

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**EuroScienceGateway (ESG)[1]** will deliver a robust, scalable, seamlessly integrated open infrastructure for data-driven research. In ESG, the **European Open Science Cloud (EOSC)[2]** initiative and its components meet with **Galaxy** as a user-oriented scientific workflow and data integration platform. Galaxy serves a broad range of scientific domains, linking computing and data resources, providing easy access for non-IT users, reproducibility features as well as various sharing options as a matter of principles. Thus, ESG fosters federated computational infrastructures and the principal demo-cratization of research data analysis.

## How does this relate to ELIXIR?

In ELIXIR, Galaxy is an integral component and has matured to a European **community[3]**, also providing centralized access to national nodes' resources and Galaxy instances. Several ELIXIR **Implementation Studies (IS)** have been carried out or are ongoing for and on Galaxy[4]. In fact, **ESG's main project parts** have been majorly supported by and grown in ELIXIR:

The **European Galaxy server** useGalaxy.eu has 70.800 registered users, who run 2 million jobs a month. On the server 331.000 workflows have been executed, resulting in 60+M jobs[5]. Support for EDAM, bio.tools, OpenEBench, WorkflowHub, RO-Crate, AAI, GA4GH standards is available, ensuring a sustainable service.

The **Pulsar Network** is a distributed job execution system linking several European data centers, scaling Galaxy across heterogeneous resources.

The **Galaxy Training Network (GTN)** is offering learning paths, TeSS integration, rich metadata annotations and training for developers, admins and scientists. 300+ contributors created >400 tutorials for 37 topics, and close to 8 years of video recordings[6].

**WorkflowHub** is an EOSC-wide registry for sharing and publishing of computational workflows, promo-ting workflows as FAIR scholarly objects by using open standards (CWL, RO-Crate, Bioschemas, GA4GH TRS)

## European Galaxy Server: Hosting (sub-)communities

The heart and soul of Galaxy are users and contributors, largely self-organized in interest groups dedicated to topics like collected below. Active onboarding of given scientific communities into the Galaxy space is a central task in ESG (WP5).

**.usegalaxy.eu**

## Project Structure

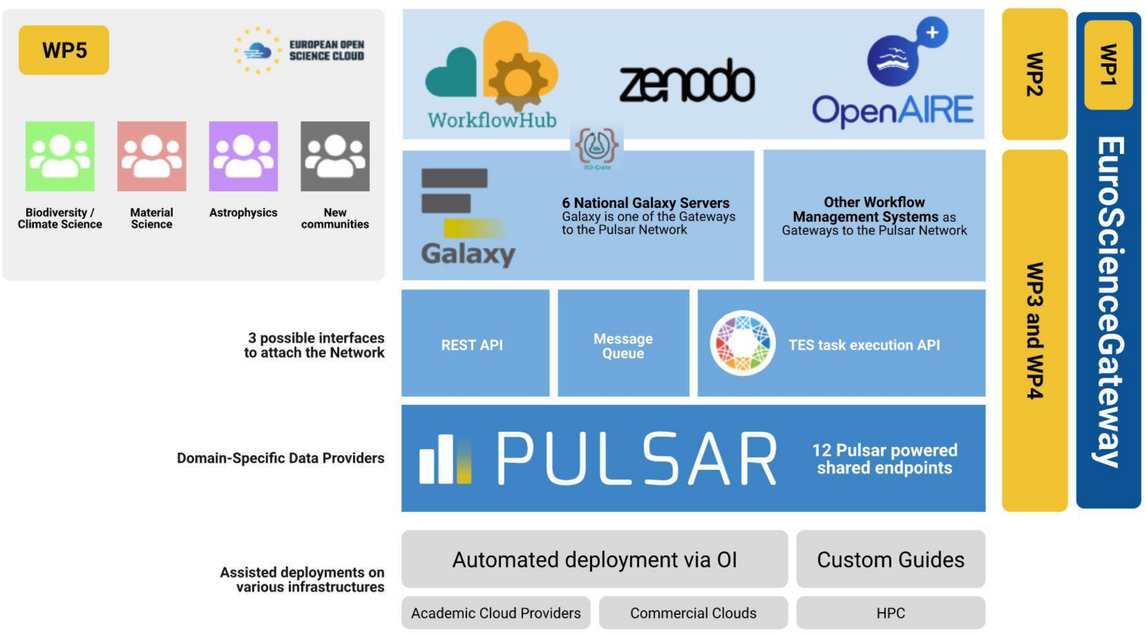


**Fig. 1** - Five work packages take care for tasks in **infrastructure** (Galaxy & PULSAR), **FAIRification** (WorkflowHub, RO-Crate) and use case-driven onboarding of **new communities** (biodiversity/climate, materials, astrophysics). Importantly, ESG builds on given infrastructures, focussing on interoperability.

## PULSAR Network: distributed computing across Europe

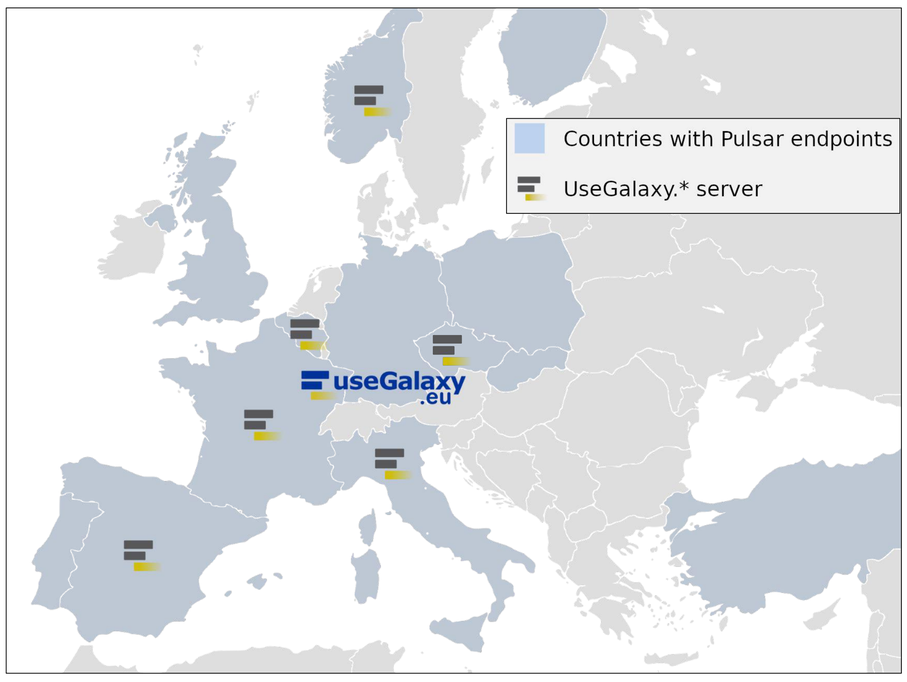


**Fig. 2** - An overall number of 18 partners from 13 countries co-operate in ESG to enable job distribution across **six national Galaxy nodes** (.* servers) and several **PULSAR endpoints**. The PULSAR project is growing, and ESG establishes **new endpoints** at partnersites in e.g. Czech Republic, Poland, Slovakia and Türkiye.

## Expected Results

**Pulsar Network**

A mature and tested (TRL-9) distributed compute network with demonstrated usage across at least 12 European partners.

**ScienceGateways**

6 national Galaxy instances operational and proven by the scientific community with more than 100,000 users.

**Communities**

Science Gateway with a custom set of tools and workflows for the Biodiversity/Climate, Materials Science and Astrophysics community.

**FAIR data**

EOSC-catalogued FAIR data and workflows that can be found, consumed, created and published by users of the ESG, demonstrated for each of the 3 domain-specific use-cases.

## ELIXIR Projects

**carried out for/on Galaxy:**

- 2018-galaxy, 2019-galaxy, 2021-galaxy
- 2018-biocontainers, 2019-containers, 2021-containerservices
- 2019-ETP1, 2022-ETP1, 2022-ETP3
- 2021-hCNVexchange, 2021-hCNVbundles
- 2021-ToolsEcosystem, 2021-IDPTools

## References

[1]https://galaxyproject.org/projects/esg/
[2]https://eosc-portal.eu/about/eosc
[3]https://elixir-europe.org/communities/galaxy
[4]https://elixir-europe.org/internal-projects/commissioned-services
[5]https://stats.galaxyproject.eu/d/000000012/galaxy-user-statistics
[6]https://training.galaxyproject.org/training-material/stats/#

## ESG Partners

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